PassFrame tuition

# Speed, Distance & Time

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

## Lesson Worksheet — Answers

### Easy / Standard

**1.** 70 km/h (350÷5=70)

**2.** B — 125×4=500

**3.** 700 km/h (2,100÷3=700)

**4.** 210 miles (70×3=210)

**5.** B — Dividing both sides by S gives T = D ÷ S

**6.** 24 km/h (84÷3.5=24)

**7.** 72 km (2 hours 15 minutes = 2.25 hours; 32×2.25=72)

**8.** 62 km/h (372÷6=62)

**9.** A, C and D (504÷7=72, 760÷8=95 and 186÷3=62 are correct; 315÷4.5 is actually 70 (not 65) and 450÷5 is actually 90 (not 99))

**10.** 90 km/h (3 hours 12 minutes = 3.2 hours; 288÷3.2=90)

**11.** 12 km/h (26.4÷2.2=12)

**12.** C — 210÷3=70

**13.** 195 km (78×2.5=195)

**14.** 80 km/h (340÷4.25=80)

**15.** 20 km/h (60 km ÷ 3 hours = 20 km/h)

### Standard

**1.** 78 km/h (585÷7.5=78)

**2.** 253 km (2 hours 45 minutes = 2.75 hours; 92×2.75=253)

**3.** C — 396÷4.4=90

**4.** 64 km (40 minutes = 2/3 hour; 96×2/3=64)

**5.** 650 km/h (3,250÷5=650)

**6.** 45 km (100 minutes = 5/3 hours; 27×5/3=45)

**7.** D — 84×3.5=294

**8.** 85 km/h (425÷5=85)

**9.** 45 km (18 minutes = 0.3 hour; 150×0.3=45)

**10.** 15.6 km (2 hours 24 minutes = 2.4 hours; 6.5×2.4=15.6)

**11.** A, B, D and E (144×2.5=360; 144×0.75=108; 144×1.25=180; 144×(1/6)=24 are correct; 20 minutes actually gives 144×(1/3)=48 km, not 50)

**12.** 70 km/h (336÷4.8=70)

**13.** 209 km (3 hours 10 minutes = 19/6 hours; 66×19/6=209)

**14.** C — 372÷4=93 km/h, the highest of the four (77, 85, 93, 88)

**15.** 80 km/h (400 km ÷ 5 hours = 80 km/h)

### Standard / Hard

**1.** 60 km/h (300 km ÷ 5 hours = 60 km/h)

**2.** C — 546÷78=7

**3.** 18 km/h (63 km ÷ 3.5 hours = 18 km/h)

**4.** 4 hours (2,760÷690=4)

**6.** C — Return time = 4 hours 48 minutes = 4.8 hours; 336÷4.8=70

**7.** 287 km (42×3.5=147, plus 56×2.5=140, total 287 km)

**8.** 595 km ((78+92)×3.5=595)

**9.** A, B, C and D (585÷65=9h; 65×4=260 km; 585÷130=4.5h;

**11.** 6 hours 15 minutes (72÷24=3 hours, plus a 45-minute rest, plus 65÷26=2.5 hours = 6.25 hours = 6 hours 15 minutes)

**12.** C — 2.5 hours × 96 km/h = 240 km

**13.** 86 km/h (645 km ÷ 7.5 hours = 86 km/h)

**14.** 73.5 km ((32−11)×3.5=73.5)

292.5÷65=4.5h — all correct; but in 6 hours the car covers 390 km, not 375 km, so that one is false)

**5.** 354 km (84×2.5=210, plus 96×1.5=144, total 354 km)

**10.** 7.25 hours (315÷90=3.5 hours, plus 315÷84=3.75 hours, total 7.25 hours)

**15.** 67.5 km/h (first part takes 450÷90=5 hours; remaining time=9−5=4 hours; 270÷4=67.5 km/h)

# Homework — Answers

## Easy / Standard

**1.** 70 km/h (315÷4.5=70)

**2.** B — Dividing both sides by T gives S = D ÷ T

**3.** 77 km (22×3.5=77)

**4.** 62 km/h (434÷7=62)

**5.** C — 250 km/h is realistic for a high-speed train

**6.** 13.2 km (5.5×2.4=13.2)

**7.** 62 km/h (279÷4.5=62)

**8.** 168 km (48×3.5=168)

**9.** A, C and D (468÷6=78, 315÷5=63 and 176÷8=22 are all correct; 245÷3.5 is actually 70 (not 68) and 592÷8 is actually 74 (not 76))

**10.** 62 km/h (372÷6=62)

**11.** 7 km/h (16.8÷2.4=7)

**12.** C — 91 km/h is the largest value

**13.** 252 km (72×3.5=252)

**14.** 72 km/h (396÷5.5=72)

**15.** 20 km/h (80 km ÷ 4 hours = 20 km/h)

## Standard

**1.** 70 km/h (483÷6.9=70)

**2.** 198 km (2 hours 15 minutes = 2.25 hours; 88×2.25=198)

**3.** C — 522÷6=87

**4.** 90 km (50 minutes = 5/6 hour; 108×5/6=90)

**5.** 680 km/h (4,080÷6=680)

**6.** 70 km (140 minutes = 7/3 hours; 30×7/3=70)

**7.** C — 76×4.5=342

**8.** 92.5 km/h (555÷6=92.5)

**9.** 66 km (24 minutes = 0.4 hour; 165×0.4=66)

**10.** 13.2 km (1 hour 50 minutes = 11/6 hours; 7.2×11/6=13.2)

**11.** A, B, D and E (156×2.5=390; 156×0.5=78; 156×1.25=195; 156×(1/6)=26 are correct; 20 minutes actually gives 156×(1/3)≈52 km, not 50)

**12.** 75 km/h (405÷5.4=75)

**13.** 238 km (2 hours 50 minutes = 17/6 hours; 84×17/6=238)

**14.** C — 494÷5.2=95 km/h, the highest of the four (92, 85, 95, 76)

**15.** 96 km/h (480 km ÷ 5 hours = 96 km/h)

## Standard / Advanced

**1.** 360 minutes (372÷62=6 hours=360 minutes)

**2.** C — 3 hours 18 minutes = 3.3 hours; 297÷3.3=90

**3.** 650 km (50 minutes = 5/6 hour; 780×5/6=650 km)

**6.** 6 hours (180÷90=2 hours, plus 260÷65=4 hours, total 6 hours)

**7.** 3 hours (combined speed 176 km/h; 528÷176=3)

**8.** A, B, D and E (550÷110=5h; 550÷125=4.4h; 110 km/h means 27.5 km every 15 minutes; 550÷88=6.25h — all correct; but in 3 hours the car

**11.** 38 km/h (total distance 285 km ÷ total time 7.5 hours = 38 km/h)

**12.** 2 hours 18 minutes (24÷16=1.5 hours plus 8÷10=0.8 hours = 2.3 hours = 2 hours 18 minutes)

**13.** C — 2 hours 30 minutes = 2.5 hours; the gap grows by 104−82=22 km/h; 22×2.5=55 km

covers 330 km, not half of 550 km (275 km), so that one is false)

**4.** 15 minutes (11÷44=0.25 hours=15 minutes)

**5.** B — Speed = 75 km/h; 375÷75=5

**9.** 8 hours (90÷30=3 hours there, 90÷18=5 hours back, total 8 hours)

**10.** C — Driving time = 315÷90=3.5 hours (3h30); plus 25-minute stop = 3h55; 06:55 + 3h55 = 10:50

**14.** 40 km/h (first part takes 400÷80=5 hours; remaining time=12−5=7 hours; 280÷7=40 km/h)

**15.** 83 km/h (leg times 1.5 + 2.25 + 1.5 = 5.25 hours; total distance 435 km; 435 ÷ 5.25 ≈ 82.9, rounded to 83 km/h)